

QQPlots and Normality Testing

MA2003B Application of Multivariate Methods
in Data Science

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Skewness, Kurtosis and QQ Plots

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In this presentation, we will explore some other distributions and show how the concepts of skewness and kurtosis can help us understand its shape.

We will also introduce QQ plots, which are useful graphical tools for comparing the distribution of a dataset against a theoretical distribution, which is usually taken to be normal.

Skewness

Definition

The **skewness** of a random variable X is defined as the third standardized moment, given by

$$\gamma_1 = \frac{\mu_3}{\sigma^3} = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3},$$

where $\mu = \mathbb{E}[X]$ is the mean and $\sigma = \sqrt{\mathbb{E}[(X - \mu)^2]}$ is the standard deviation of X .

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Skewness measures the asymmetry of a probability distribution.

A distribution is said to be **positively skewed** (or right-skewed) if it has a longer tail on the right side, and **negatively skewed** (or left-skewed) if it has a longer tail on the left side.

Gamma Distribution

Definition

Let X be a continuous random variable following a **Gamma distribution** with shape parameter $\alpha > 0$ and rate parameter $\beta > 0$. The probability density function (PDF) of X is given by

$$f_X(x; \alpha, \beta) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & x > 0, \\ 0, & x \leq 0, \end{cases} \quad (1)$$

where $\Gamma(\alpha)$ denotes the Gamma function, defined as

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt.$$

The parameters α and β control the shape and rate of the distribution, respectively.

Among other applications, the Gamma distribution is frequently used to model the number of total insurance claims over a given time period.

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Remark

The skewness of a Gamma distribution is given by

$$\gamma_1 = \frac{2}{\sqrt{\alpha}}.$$

This means that the skewness is always positive, and it approaches zero as α increases. In other words, the distribution becomes more symmetric as α becomes larger.

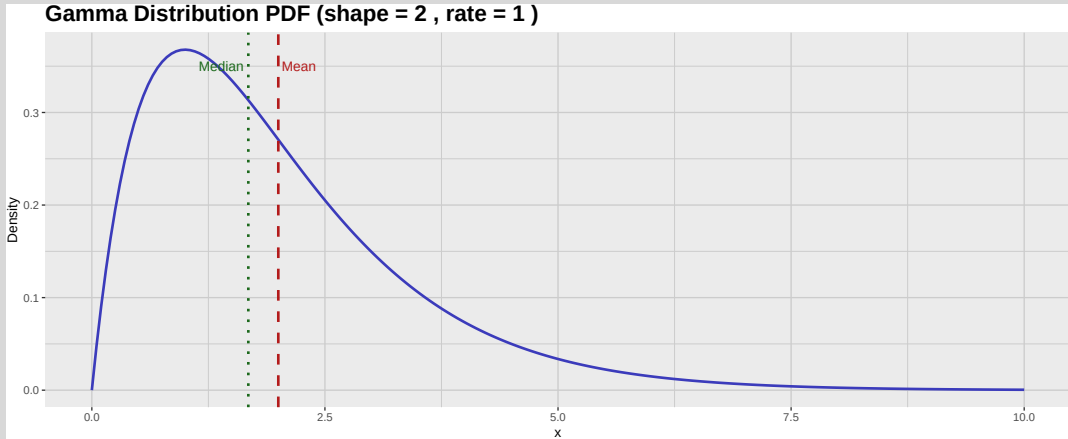


Figure 1: PDF of the gamma distribution with mean and median highlighted.

Kurtosis

Definition

The **kurtosis** of a random variable X is defined as the fourth standardized moment, given by

$$\gamma_2 = \frac{\mu_4}{\sigma^4} = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4},$$

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A distribution with high kurtosis has heavier tails and a sharper peak than a normal distribution, while a distribution with low kurtosis has lighter tails and a flatter peak.

The kurtosis of a normal distribution is 3, and it is often adjusted to be 0 for comparison purposes.

Laplace Distribution

Definition

Let X be a continuous random variable following a **Laplace distribution** with location parameter $\mu \in \mathbb{R}$ and scale parameter $b > 0$. The probability density function (PDF) of X is given by

$$f_X(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right), \quad x \in \mathbb{R}. \quad (2)$$

The parameters μ and b control the location and scale of the distribution, respectively.

Among other applications, the Laplace distribution is often used in image processing and computer vision, particularly in modeling noise in images.

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Remark

The kurtosis of a Laplace distribution is given by

$$\gamma_2 = 6.$$

This means that the Laplace distribution has heavier tails than a normal distribution, which has a kurtosis of 3.

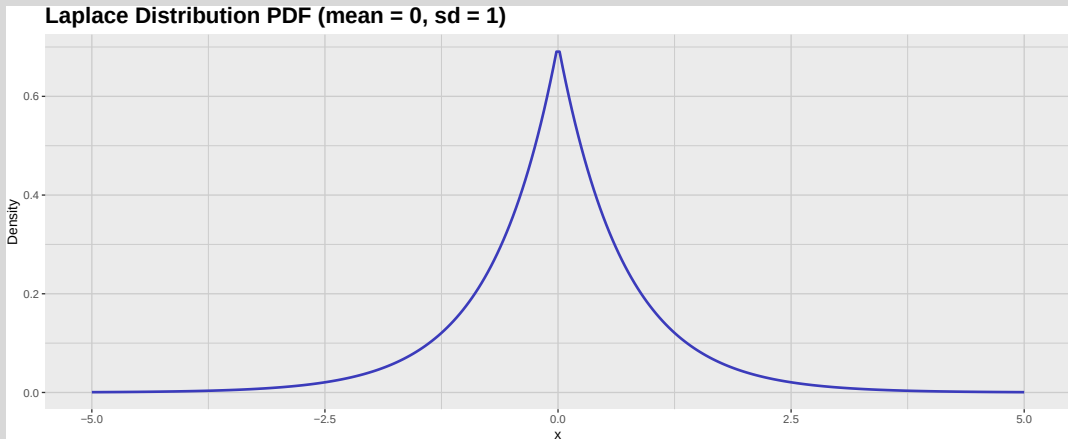


Figure 2: PDF of the Laplace distribution with mean zero and variance one.

QQ Plots

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They are particularly useful for assessing whether a dataset follows a normal distribution.

Constructing a QQ Plot

To construct a QQ plot, you should follow these steps:

1. **Sort the sample data:** Arrange the data in increasing order:

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}.$$

2. **Calculate plotting positions (empirical quantiles):** For each ordered value $x_{(i)}$, compute the corresponding probability

$$p_i = \frac{i - 0.5}{n}, \quad i = 1, 2, \dots, n.$$

Constructing a QQ Plot

3. **Find theoretical quantiles of the normal distribution:** Compute the theoretical quantiles from the inverse cumulative distribution function (CDF) of the normal distribution:

$$q_i = \Phi^{-1}(p_i),$$

where Φ^{-1} is the quantile function of the standard normal distribution.

4. **Plot the points:** Plot the pairs $(q_i, x_{(i)})$ with q_i on the x-axis and $x_{(i)}$ on the y-axis.
5. **Add a reference line:** Typically, we add a line through the first and third quartiles to help assess normality.

The closer the points lie to this line, the more the sample resembles the specified normal distribution.

Gamma Distribution: QQ Plot and Histogram

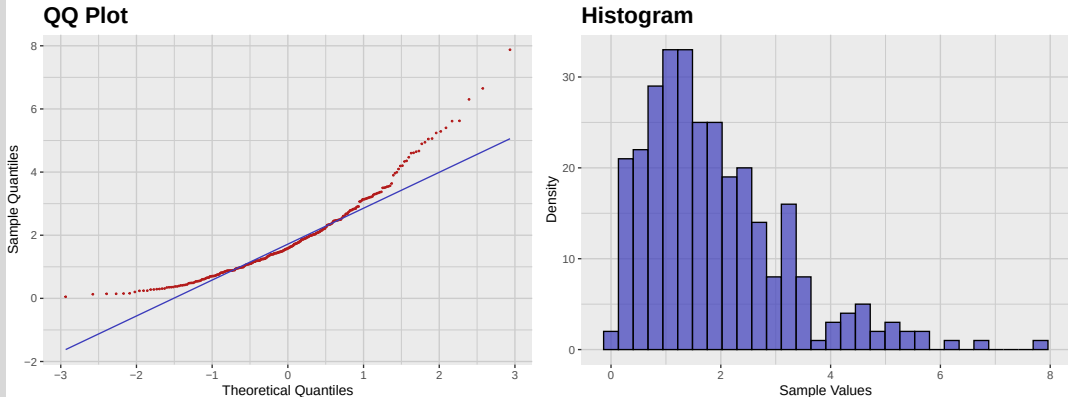


Figure 3: QQ plot and histogram of the Gamma distribution samples.

Laplace Distribution: QQ Plot and Histogram

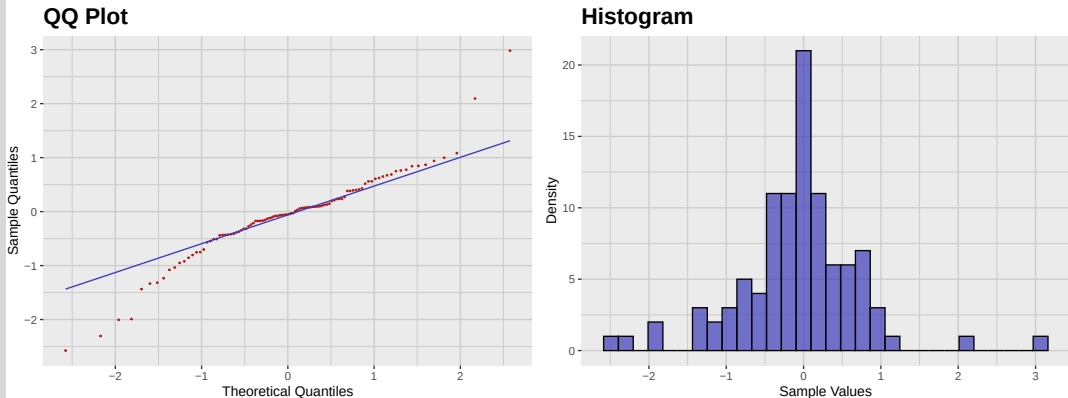


Figure 4: QQ plot and histogram of the Laplace distribution samples.